H5 — ML + Embedded Track

Member: ______

Your role

You own the Colab runbook. You support every training run during the campaign. You keep the tracking sheet current.

Weekly tasks

Week 1 — Colab probe + runbook

- Task: Run a 50-step probe of train.py on Colab. Test checkpoint save, session kill, and resume. Write the Colab runbook.
- Deliverable: The Colab runbook in the club repo.
- Verify: A killed session resumes from the last checkpoint.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Run support

- Task: Watch all 6 training runs. Fix Colab issues. Keep the tracking sheet current.
- Deliverable: The daily run status log.
- Verify: Every run has a status row in the tracking sheet.
- Learn: Embeddings and next-token prediction. Explain why checkpoints save to Drive.

Week 3 — Tracking updates

- Task: Update the tracking sheet with every week-3 number. Flag every gap.
- Deliverable: A complete tracking sheet.
- Verify: R2 uses your numbers in the report without edits.
- Learn: The transformer block. Explain the training loop with the loss curve.

Week 4 — Report consolidation

- Task: Consolidate the tracking sheet into the verification report.
- Deliverable: The consolidated results table.
- Verify: The table matches the tracking sheet row by row.
- Learn: Sampling and generation. Explain why the seed fixes the sampling stream.

Week 5 — Decision criteria

- Task: Write the use-case decision criteria with the team. Effort, impact, demo-ability.
- Deliverable: The decision criteria doc.
- Verify: Every member votes with the same criteria.
- Learn: RoPE and KV cache. Draw the cache growth over 200 tokens.

Week 6 — Decision presentation

- Task: Present the use-case decision to the team. Collect feedback.
- Deliverable: The feedback summary.
- Verify: The final decision appears in USE_CASE_DECISION.md.
- Learn: Fine-tuning. Explain what a use case needs from a TinyStories-class model.

Week 7 — Teach memory tiers

- Task: Present the memory tiers to the team at the meetup. SRAM, PSRAM, flash.
- Deliverable: Your teach-back slides.
- Verify: E2 can explain why the 25M table lives in flash.
- Learn: Prepare your presentation deeply.

Week 8 — Member feedback

- Task: Collect member feedback on the project and the tasks.
- Deliverable: The feedback summary in the club repo.
- Verify: Every member submits one form.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: https://github.com/slvDev/esp32-ai
- Colab: https://colab.research.google.com
- Kaggle: https://www.kaggle.com
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi